Calculus Basics

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1 Continuity

We start with the concept of continuity. A function $f : \mathbb{R} \to \mathbb{R}$ is said to be continuous at a point $x = a$ if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all x within the interval $(a-\delta, a+\delta)$, the value of $f(x)$ lies within the interval $(f(a) - \epsilon, f(a) + \epsilon)$. Mathematically, this can be written as:

$$\forall \epsilon > 0,\, \exists \delta > 0 \,\text{such that}\, |x - a| < \delta \implies |f(x) - f(a)| < \epsilon.$$

2 Lipschitz continuous

For most of this course, we will focus on "nice" functions, which are continuous and possibly even smoother. In particular, we will discuss Lipschitz continuous functions. A function f is said to be Lipschitz continuous if there exists a constant $L \geq 0$ such that for all $x, y \in \mathbb{R}^n$,

$$|f(x) - f(y)| \leq L\|x - y\|. \tag{1}$$

This relation implies that the difference in function values is controlled by how close the input points x and y are. For instance, linear functions are Lipschitz continuous, but quadratic functions are not. One key consequence of Lipschitz continuity is that the gradient of the function is bounded by the constant L.

3 Hölder Continuity

We can generalize the notion of Lipschitz continuity by introducing the concept of Hölder continuity. A function f is said to be Hölder continuous with exponent $\alpha \in [0, 1]$ if there exists a constant $C \geq 0$ such that for all $x, y \in \mathbb{R}^n$,

$$|f(x) - f(y)| \leq C\|x - y\|^{\alpha}. \tag{2}$$

When $\alpha = 1$, we recover the definition of Lipschitz continuity. When $\alpha = 0$, the function itself is bounded. While we will primarily work with Lipschitz continuous functions, it is worth noting that some functions may be Hölder continuous with an exponent α strictly less than one.

4 Gradient and Hessian

Next, we review the concepts of gradient and Hessian, which generalize first-order and second-order derivatives, respectively. Consider a function $f : \mathbb{R}^d \to \mathbb{R}$. The gradient of f, denoted by $\nabla f(x)$, is a column vector in $\mathbb{R}^d$ where the i-th entry is the partial derivative of f with respect to the i-th variable x_i given by:

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{pmatrix} \in \mathbb{R}^d.$$

The Hessian of f, denoted by $H_f(x)$, is a $d \times d$ matrix, symmetric by nature, where the (i, j)-th entry is the second partial derivative of f with respect to x_i and x_j given by:

$$\nabla^2 f(x) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_d} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_d} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_d \partial x_1} & \frac{\partial^2 f}{\partial x_d \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_d^2} \end{pmatrix} \in \mathbb{R}^{d \times d}.$$

4.1 Example: Quadratic Function

Consider the quadratic function

$$f(x) = \frac{1}{2} x^\top Q x + q^\top x + c, \tag{3}$$

where $x \in \mathbb{R}^d$, Q is a symmetric matrix in $\mathbb{R}^{d \times d}$, $q \in \mathbb{R}^d$, and $c \in \mathbb{R}$. The gradient and Hessian of this function are given by

$$\nabla f(x) = Qx + q, \tag{4}$$

and

$$H_f(x) = Q, \tag{5}$$

respectively. If Q is not symmetric, in the gradient and Hessian, Q is replaced by $(Q + Q^\top)/2$.

5 Determinant and Its Gradient

Although we will not focus heavily on the concept of the determinant in this course, it is useful to have a reference point. The determinant of a matrix and its gradient might appear in applications such as maximum likelihood estimation, where one may need to compute the logarithm of the determinant of a Hessian matrix.

5.1 Determinant Formula

The determinant of A can be computed using the cofactor expansion along any row or column. For example, expanding along the j-th column:

$$|A| = \sum_{i=1}^{n} a_{ij}(-1)^{i+j}|A_{-i,-j}|,$$

where $A_{-i,-j}$ denotes the $(n-1) \times (n-1)$ submatrix of A obtained by deleting the i-th row and j-th column.

5.2 Gradient of the Determinant

The gradient of a function $f(M)$ with respect to the matrix M is given by:

$$\nabla_M f(M) = \left[\frac{\partial f}{\partial M_{ij}}\right].$$

For the determinant, the gradient is related to the adjoint matrix:

$$\nabla|A| = \text{adj}(A)^T = |A|(A^T)^{-1},$$

where the adjoint of A is the transpose of the cofactor matrix.

5.3 Gradient of the Log-Determinant

The gradient of the log-determinant function can be derived as follows:

$$\nabla \log\det(A) = \nabla(\log|A|) = \frac{1}{|A|}\nabla|A| = \frac{1}{|A|}\text{adj}(A)^T.$$

Since $\text{adj}(A)^T = |A|(A^T)^{-1}$, we have:

$$\nabla \log\det(A) = \frac{1}{|A|}|A|(A^T)^{-1} = (A^T)^{-1} = A^{-1}.$$

6 Chain Rule

The chain rule is a fundamental tool for differentiating compositions of functions. Suppose $g : \mathbb{R}^d \to \mathbb{R}^n$ and $f : \mathbb{R}^n \to \mathbb{R}$. Then, the gradient of the composition $h(x) = f(g(x))$ is given by

$$\nabla h(x) = \nabla g(x) \nabla f(g(x)), \tag{6}$$

where $\nabla g(x)$ is a $d \times n$ matrix, and $\nabla f(g(x))$ is a column vector in $\mathbb{R}^n$. Therefore, $\nabla h(x)$ is a column vector in $\mathbb{R}^d$.

6.1 Example: Gradient of a Composed Function

Consider the function

$$f(x) = \frac{1}{2}\|x - h(x)\|_2^2, \tag{7}$$

where $h : \mathbb{R}^d \to \mathbb{R}^d$. The gradient of $f(x)$ is given by

$$\nabla f(x) = (I - \nabla h(x))(x - h(x)). \tag{8}$$

For instance, if $h(x) = Ax - b$, where A is a matrix, then $\nabla h(x) = A^\top$, and the gradient becomes

$$\nabla f(x) = A^\top(Ax - b). \tag{9}$$

Similarly, if $h(x) = Ax - x$, then:

$$\nabla_x h(x) = A^T - I.$$

Thus, the gradient $\nabla f(x)$ becomes:

$$\nabla f(x) = (I - (A^T - I))(x - h(x)) = A^T Ax.$$

7 Mean Value Theorem

The Mean Value Theorem (MVT) states that if a function f is continuously differentiable, there exists a constant $\alpha \in [0, 1]$ such that for any $x, y \in \mathbb{R}^d$,

$$f(y) = f(x) + \nabla f(x + \alpha(y - x))^\top (y - x). \tag{10}$$

If $\alpha = 0$, the right-hand side becomes a linear function of y, which is useful in analyzing optimization algorithms for deep learning models. The second part of the MVT states that if f is twice differentiable, then

$$f(y) = f(x) + \nabla f(x)^\top (y - x) + \frac{1}{2}(y - x)^\top H_f(\xi)(y - x), \tag{11}$$

where ξ lies on the line segment joining x and y. This result is crucial for proving smoothness properties of functions.

8 Taylor Approximation

The Taylor approximation states that if f is differentiable, then for y close to x,

$$f(y) \approx f(x) + \nabla f(x)^\top (y - x). \tag{12}$$

This approximation helps simplify the analysis of optimization problems since linear functions are easier to work with. If second-order information is included, the Taylor expansion becomes

$$f(y) \approx f(x) + \nabla f(x)^\top (y - x) + \frac{1}{2}(y - x)^\top H_f(x)(y - x), \tag{13}$$

which provides a tighter approximation.

9 Fundamental Theorem of Calculus

Finally, the Fundamental Theorem of Calculus in multiple dimensions states that for a function F defined as the integral of its gradient,

$$F(x) = \int \nabla F(x) \cdot dx. \tag{14}$$

This theorem extends the familiar univariate version to higher dimensions, where the dot product is used. This theorem is used in proof of many smoothness properties that we may discuss in this course, but may omit their prrofs.